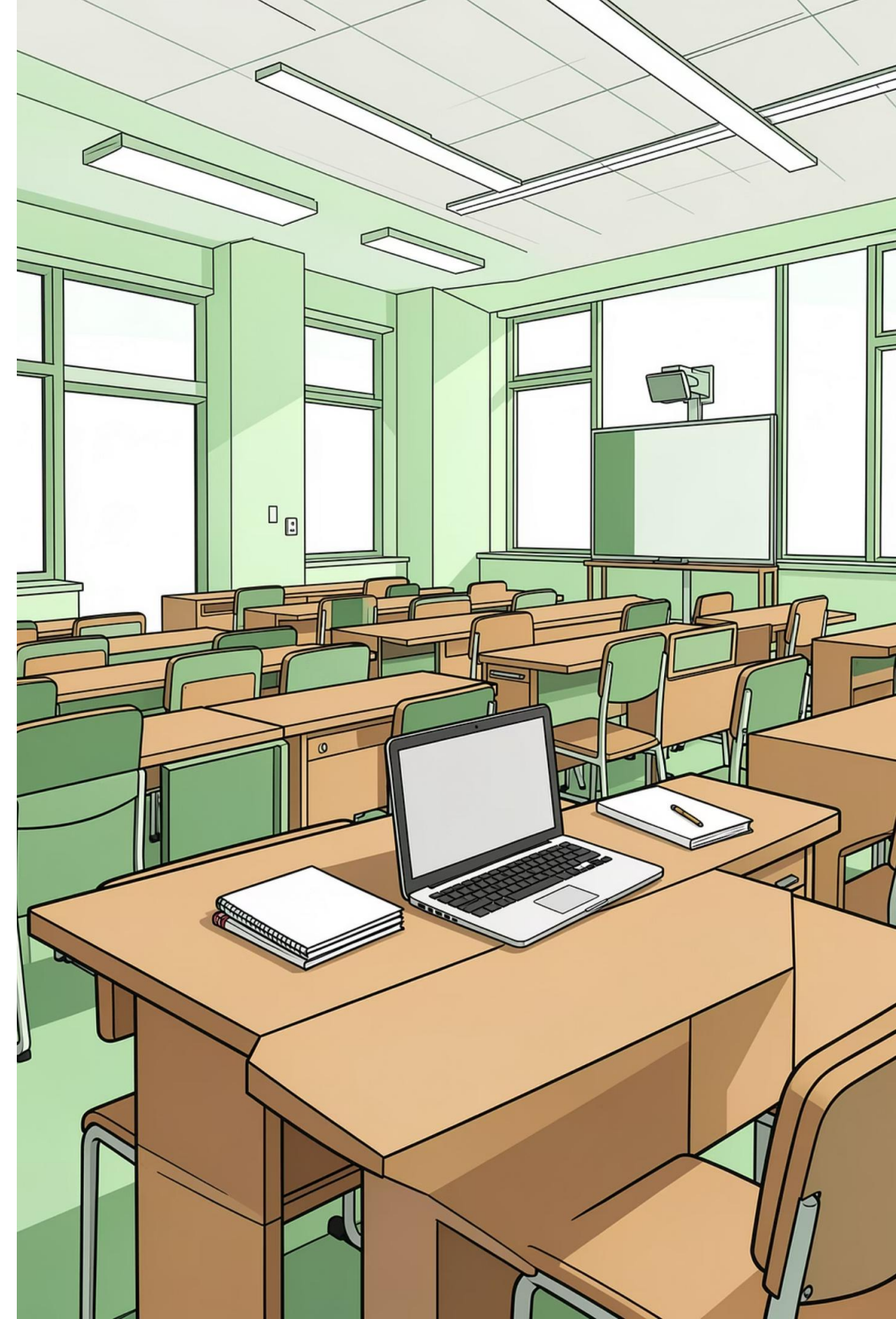


# Online Billing System

Student Names: (Add Names) · Guide: (Guide Name) · Department of  
Computer Science · College Name : MBIT · Academic Year: 2023–24



# Introduction

- Importance: billing systems underpin revenue, compliance, and customer trust
- Problems with manual billing: slow processing, human errors, paper dependency
- Need for automation: speed, accuracy, centralized records
- Solution overview: web-based system for invoicing, tracking, and reporting





# Literature Review

- Existing systems: packaged POS, cloud invoicing platforms, ERP modules
- Benefits of computerized billing: faster invoicing, audit trails, analytics
- Database-driven apps: reliable persistence, ACID transactions, query-based reporting
- Limitations of traditional systems: poor scalability, fragmented records, limited access





# Problem Statement

- Manual billing causes operational delays and longer customer wait times
- Calculation errors and inconsistent pricing lead to disputes
- Poor record management: lost invoices, difficult audits
- Requirement: centralized, automated online billing to ensure accuracy and accessibility



## Hardware & Software Requirements

### Hardware:

- Computer system — minimum 4GB RAM, stable Internet connection

### Software:

- Programming language: Python / Java / PHP (choose per team)
- Frontend: HTML, CSS, JavaScript (responsive UI)
- Database: MySQL (relational storage, transactions)
- IDE & server: VS Code / IntelliJ / XAMPP / Apache or similar

# System Architecture



Architecture emphasizes separation of concerns: Presentation (UI), Application logic (billing workflows), Data layer (MySQL), and an invoice generator for PDF/print outputs.

# UML Diagrams (Overview)

1

## Use Case Diagram

Actors: Admin, Cashier, Customer — use cases: Login, Create Invoice, Manage Products, Generate Reports

2

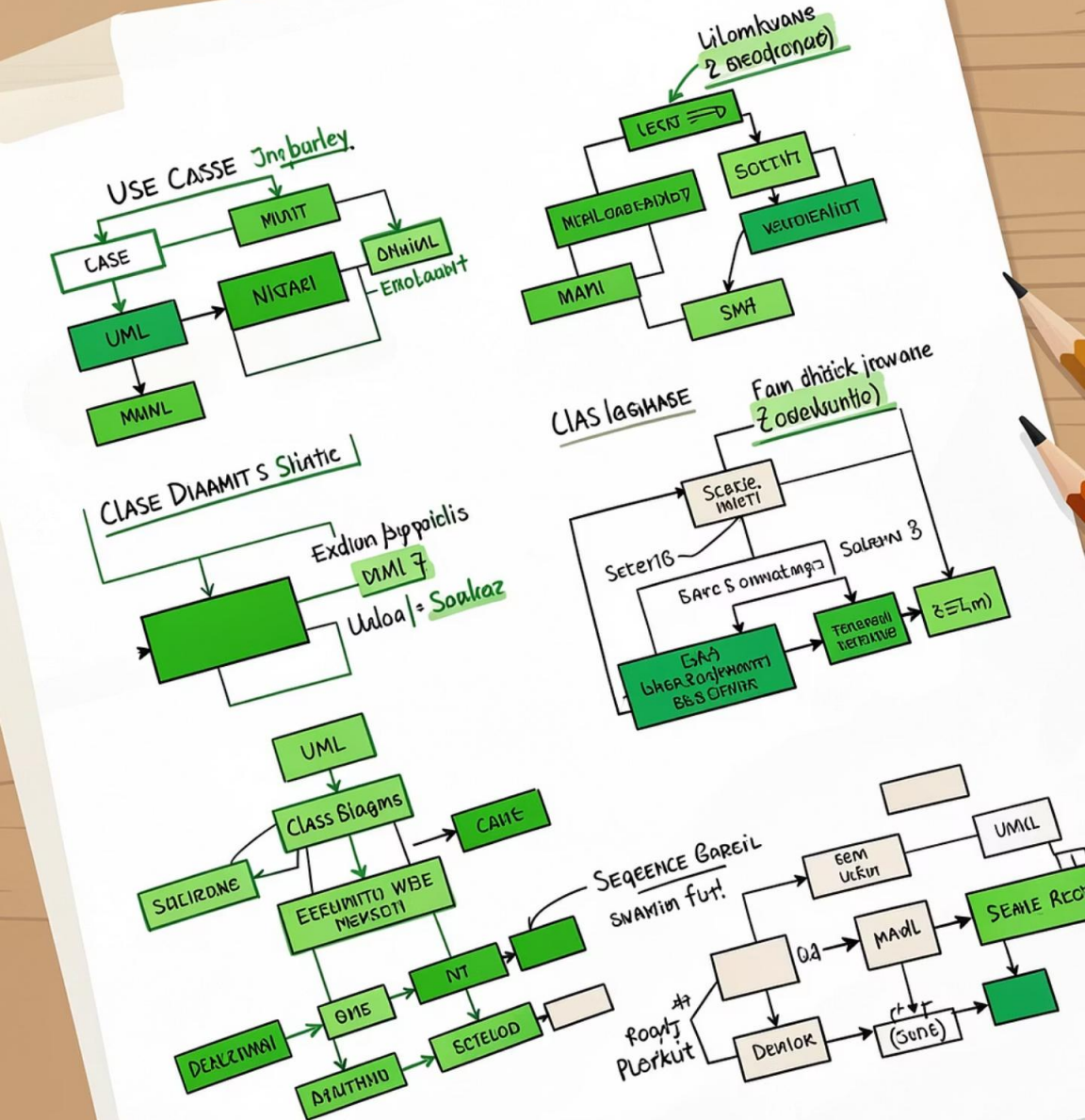
## Class Diagram

Core classes: User, Product, Invoice, LineItem, Payment, Report — relationships and key attributes shown

3

## Sequence Diagram

Example flow: Cashier → Create Invoice → Billing Module → Database → Invoice Generation → Print/Email



# Implementation — Screenshots (placeholders)



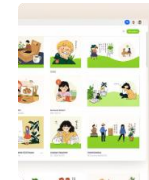
## Login Page

Secure authentication, role-based access control



## Dashboard

At-a-glance KPIs, recent activity, shortcuts



## Product Management

Add/edit items, pricing, stock tracking



## Billing & Invoice Screen

Line items, tax/discount calculations, PDF export

Replace mockups with actual screenshots before final submission.



# Future Work & Conclusion

- Future work: integrate online payments (gateway APIs), cloud storage for backups, and native mobile app
- Scalability: move to microservices and horizontal DB scaling for high-volume usage
- Security: TLS, input validation, role-based authorization, audit logging
- Conclusion: system improves efficiency, reduces errors, and secures digital records for audits



# References

- Core Texts
  - Software Engineering (standard textbooks)
  - Database System Concepts — Silberschatz et al.
- Documentation & Articles
  - MySQL official documentation
  - Research articles on e-invoicing and billing automation
- Tools & Libraries
  - Frontend: HTML/CSS/JavaScript frameworks
  - Backend: chosen language docs (Python/Java/PHP), server stacks

Acknowledgements: guide, department staff, and peers for feedback during development and testing.